

QM II Homework Assignment 8

Time Dependent Perturbation Theory:
The Sudden and Adiabatic Approximations

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1. Shankar 18.2.3.

Consider a particle in the ground state of a box of length L . Argue on semi-classical grounds that the natural time period associated with it is $T \approx nL^2/\hbar\pi$. If the box expands symmetrically to double its size in time $\tau \ll T$ what is the probability of catching the particle in the ground state of the new box?

The relative semi-classical time scale is the time that it would take a particle to cross the well. The quantum mechanical ground state energy of the infinite square well is

$$E_1 = \frac{\hbar^2 \pi^2}{2mL^2}.$$

The classical speed of a particle with this energy is

$$v = \sqrt{\frac{2E}{m}} = \frac{\hbar\pi}{mL}.$$

The time to travel across the well is therefore

$$T = \frac{v}{L} = \boxed{\frac{mL^2}{\hbar\pi}}$$

We can use this time scale to judge whether or not a perturbation to the potential is sudden or adiabatic. For changes on much shorter timescales, the classical particle will not have crossed the well and will therefore not know about the change. On the other hand, for changes much longer than this timescale the particle will bounce off of the walls many times during the change and will therefore not gain or lose energy during the perturbation.

For this problem, we are given that the time over which the perturbation happens is much less than our timescale; $\tau \ll T$. This is therefore a sudden perturbation and we will expect the wavefunction of state to be unchanged during the course of the perturbation. The initial (and therefore final) state of the particle is

$$\psi(x) = \sqrt{\frac{2}{L}} \cos\left(\frac{\pi x}{L}\right).$$

The new ground state is

$$\psi_0(x) = \sqrt{\frac{1}{L}} \cos\left(\frac{\pi x}{2L}\right).$$

Hence, the probability of finding the particle in the ground state is

$$\begin{aligned}
 P(\psi_0) &= \left| \int_{-L/2}^{L/2} \sqrt{\frac{2}{L^2}} \cos\left(\frac{\pi x}{2L}\right) \cos\left(\frac{\pi x}{L}\right) dx \right|^2 \\
 &= \left| \frac{\sqrt{2}}{L} \left(\frac{4\sqrt{2}l}{3\pi} \right) \right|^2 \\
 &= \left(\frac{8}{3\pi} \right)^2 \approx 0.721.
 \end{aligned}$$

2. Shankar 18.2.5

An oscillator is in the ground state of $\mathcal{H} = \mathcal{H}^0 + \mathcal{H}^1$, where the time-independent perturbation $\mathcal{H}^1$ is the linear potential $V = -fx$. If at $t = 0$, $\mathcal{H}^1$ is abruptly turned off, show that the probability that the system is in the n th eigenstate of $\mathcal{H}^0$ is given by the Poisson distribution

$$P(n) = \frac{e^{-\lambda} \lambda^n}{n!} \quad \text{where} \quad \lambda = \frac{f^2}{2m\omega^3\hbar}$$

We need to find the eigenstates of the perturbed Hamiltonian before we can proceed with the problem. We are fortunate in that this system is exactly solvable. Consider the Hamiltonian,

$$\begin{aligned}
 \mathcal{H} &= \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2 - fx \\
 &= \frac{p^2}{2m} + \frac{1}{2}m\omega^2 \left(x - \frac{f}{m\omega^2} \right)^2 - \frac{1}{2} \frac{f^2}{m\omega^2}
 \end{aligned}$$

From this form we can see that the perturbed Hamiltonian is simply a harmonic oscillator with a shifted constant energy and a translated center. The perturbed ground state wavefunction can be found from the old unperturbed one by applying the translation operator.

$$\begin{aligned}
 |0\rangle &= T\left(\frac{f}{m\omega^2}\right) |0^0\rangle \\
 &= \exp\left(-i \frac{f}{\hbar m\omega^2} p\right) |0^0\rangle \\
 &= \exp\left(-i \frac{f}{\hbar m\omega^2} i \sqrt{\frac{\hbar m\omega}{2}} (a^\dagger - a)\right) |0^0\rangle \\
 &= \exp\left(\sqrt{\frac{f^2}{2m\hbar\omega^3}} (a^\dagger - a)\right) |0^0\rangle
 \end{aligned}$$

We want to expand the exponential into two terms, but we must be careful because non-commuting operators do not satisfy the usual relation ship. Recall that for two operators A and B ,

$$e^{A+B} = e^A e^B e^{-\frac{1}{2}[A, B]}.$$

Before we return to the above equation, let us also compute a useful commutator. For operators A and B and a scalar quantity a ,

$$[aA, -aB] = (aA)(-aB) - (-aB)(aA) = a^2(-AB + BA) = a^2[B, A].$$

Applying these two relations, the new state becomes

$$\begin{aligned} |0\rangle &= \exp\left(\sqrt{\frac{f^2}{2m\hbar\omega^3}}a^\dagger\right)\exp\left(-\sqrt{\frac{f^2}{2m\hbar\omega^3}}a\right)\exp\left(-\frac{f^2}{4m\hbar\omega^2}\right)|0^0\rangle \\ &= e^{\sqrt{\lambda}a^\dagger}e^{-\sqrt{\lambda}a}e^{-\frac{1}{2}\lambda}|0^0\rangle \end{aligned}$$

We now want to expand the two exponential terms in terms of their Taylor expansions. Since the operators will be acting on the ground state, all of the terms in the expansion of the second operator will go to zero except for the identity. We can therefore drop it completely. Before we expand the other term, it is helpful to recall that the eigenstates of the harmonic oscillator can be expressed in terms of the ground state and the raising operator as

$$|n^0\rangle = \frac{(a^\dagger)^n}{\sqrt{n!}}|0^0\rangle$$

Lets expand the exponential in the above expression.

$$\begin{aligned} |0\rangle &= e^{-\frac{1}{2}\lambda}\frac{(\sqrt{\lambda})^n}{n!}(a^\dagger)^n|0^0\rangle \\ &= e^{-\frac{1}{2}\lambda}\frac{(\sqrt{\lambda})^n}{\sqrt{n!}}|n^0\rangle \end{aligned}$$

Finally, since this perturbation is applied suddenly, the system will remain in this state even after the perturbation is applied. The probability of being in one of the new eigenstates of the unperturbed harmonic oscillator is simply its overlap with those states.

$$\begin{aligned} P(n) &= |\langle n^0|0\rangle|^2 \\ &= \boxed{\frac{e^{-\lambda}\lambda^n}{n!}} \end{aligned}$$

3. For a spin $\frac{1}{2}$ particle, consider the time dependent Hamiltonian,

$$\mathcal{H}(t) = -\hbar\omega\hat{n}(t) \cdot \vec{\sigma},$$

where $\vec{\sigma}$ is the vector formed from the Pauli spin matrices. The unit vector $\hat{n}$ changes with time. Suppose initially $\hat{n} = \hat{z}$, and the spin is in the ground state. Next, $\hat{n}$ is rotated so that it points in the $\hat{x}$ direction. What is the wavefunction when $\hat{n}$ reaches $\hat{x}$ for the case of (a) a rotation in time $T \ll \omega^{-1}$ and (b) a rotation in time $T \gg \omega^{-1}$? (Note: you do not need to have the correct phase for the wave function for case (b). The phase will depend on the exact details of how the state is changed from $\hat{z}$ to $\hat{x}$ and we will discuss this later.) For both cases, what is the probability that the spin will be in the new ground state ($\hat{n} = \hat{x}$)?

Lets begin this problem by determining the eigenvalues and eigenvectors of the initial and final Hamiltonians. At time $t = 0$, $\hat{n}$ points in the $\hat{z}$ direction and the Hamiltonian is

$$\mathcal{H}(0) = -\hbar\omega\sigma_z = \begin{pmatrix} -\hbar\omega & 0 \\ 0 & \hbar\omega \end{pmatrix}.$$

This is clearly already diagonal and so we can identify the eigenvectors and their energies as

$$|-\hbar\omega\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \& \quad |\hbar\omega\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Now for the final Hamiltonian. At time $t = T$, $\hat{n}$ points in the $\hat{x}$ direction and the Hamiltonian has the form

$$\mathcal{H}(T) = -\hbar\omega\sigma_x = \begin{pmatrix} 0 & -\hbar\omega \\ -\hbar\omega & 0 \end{pmatrix}.$$

We diagonalize this in the usual way, by calculating the characteristic determinant and setting it equal to zero.

$$\begin{vmatrix} -\lambda & -\hbar\omega \\ -\hbar\omega & -\lambda \end{vmatrix} = \lambda^2 - \hbar^2\omega^2 = (\lambda - \hbar\omega)(\lambda + \hbar\omega) = 0 \quad \Rightarrow \quad \lambda = \pm\hbar\omega$$

Plugging these eigenvalues back in to find the eigenvectors reveals them to be

$$|-\hbar\omega\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \& \quad |\hbar\omega\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Part a

Since the perturbation in this case is very sudden, we expect the wavefunction to remain unchanged after the perturbation is applied. Hence, the perturbed wavefunction is

$$\boxed{|\psi_{\hat{x}}\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}}.$$

Hence, the probability of it being in the new ground state is

$$P(-\hbar\omega) = \left| \begin{pmatrix} 1 & 0 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right|^2 = \boxed{\frac{1}{2}}.$$

Part b

In this case, the approximation is adiabatic and we will expect the state of σ_z to track to the ground state of σ_x . Hence, the state after the perturbation will be

$$\boxed{|\psi_{\hat{x}}\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}}.$$

Which means that the probability of finding the system in the ground state is one.

$$P(-\hbar\omega) = \left| \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right|^2 = \boxed{1}$$

4. A time dependent Hamiltonian has the form:

$$\mathcal{H}(t) = \frac{p^2}{2m} + \frac{1}{2}m\omega^2(t)x^2,$$

where the only time dependence comes from $\omega(t)$. This is equivalent to varying the spring constant with time. In this problem, we will make use of equations 7.3.21 and 7.3.22 from the book for the harmonic oscillator wavefunctions, together with the Gaussian integrals:

$$\int_{-\infty}^{\infty} dx e^{-ax^2} = \sqrt{\frac{\pi}{a}}$$

$$\int_{-\infty}^{\infty} dx x^2 e^{-ax^2} = \frac{1}{2}\sqrt{\pi a^3}$$

- (a) From $t = 0$ to $t = T$ the angular frequency changes from ω_0 to $2\omega_0$. At $t = 0$ the system is in the ground state of $\mathcal{H}(t = 0)$. If $\omega_0 T \ll 1$, what is the probability that the system will still be in the ground state at $t = T$?
- (b) What is the probability that it will be in the first excited state at $t = T$? What is the probability that it will be in the second excited state at $t = T$?
- (c) Suppose instead that $\omega_0 T \gg 1$. What is the probability that the system will still be in the ground state at $t = T$?
- (d) What is the probability that it will be in the first excited state at $t = T$? What is the probability that it will be in the second excited state at $t = T$?

Part a

Recall that the wavefunctions of the harmonic oscillator are Gaussians modulated by Hermite polynomials. The ground state is

$$|\psi_0(\omega_0, x)\rangle = \left(\frac{m\omega_0}{\pi\hbar}\right)^{\frac{1}{4}} \exp\left(-\frac{m\omega_0 x^2}{2\hbar}\right).$$

Since the semi-classical time scale of the harmonic oscillator is $T = \frac{1}{f} \sim \frac{1}{\omega}$ and this perturbation happens on a time scale much less than this, the wavefunction will remain unchanged through the perturbation. Hence, the probability of the particle being in the new ground state is simply the overlap with the wavefunction of the new ground state.

$$\begin{aligned} P(|\psi_0(2\omega_0, x)\rangle) &= |\langle\psi_0(\omega_0, x)|\psi_0(2\omega_0, x)\rangle|^2 \\ &= \left[\int_{-\infty}^{\infty} dx \sqrt{\frac{\sqrt{2}m\omega_0}{\pi\hbar}} \exp\left(-\frac{3m\omega_0 x^2}{\hbar}\right) \right]^2 \\ &= \frac{\sqrt{2}m\omega_0}{\pi\hbar} \frac{2\hbar\pi}{3m\omega_0} \\ &= \left[\frac{2^{\frac{3}{2}}}{3} \right] \approx 0.9428 \end{aligned}$$

Part b

Since this is the same situation as in part a, the system will remain in the original ground state after the perturbation is applied. The probability of being in the first excited state of the

perturbed potential is

$$P(|\psi_1(2\omega_0, x)\rangle) = \left[\frac{m\omega_0}{\hbar} \frac{2^{\frac{5}{4}}}{\sqrt{\pi}} \int_{-\infty}^{\infty} dx x \exp\left(-\frac{m\omega_0 x^2}{2\hbar}\right) \right]^2 = \boxed{0}.$$

So, the particle has a zero probability of ending up in the first excited state of the perturbed potential.

For the second excited state,

$$\begin{aligned} P(|\psi_2(2\omega_0, x)\rangle) &= \left[\sqrt{\frac{m\omega_0}{\pi\hbar\sqrt{2}}} \int_{-\infty}^{\infty} dx \left(\frac{4m\omega_0}{\hbar} x^2 - 1 \right) \exp\left(-\frac{3m\omega_0 x^2}{2\hbar}\right) \right]^2 \\ &= \frac{m\omega_0}{\pi\hbar\sqrt{2}} \left[\sqrt{\frac{2\hbar}{3m\omega_0}} \int_{-\infty}^{\infty} dx \left(\frac{8}{3} u^2 - 1 \right) e^{-u^2} du \right]^2 \\ &= \frac{\sqrt{2}}{3\pi} \left[\frac{8}{3} \frac{\sqrt{\pi}}{2} - \sqrt{\pi} \right]^2 \\ &= \boxed{\frac{\sqrt{2}}{27}} \approx 0.0524 \end{aligned}$$

So we see that the particle is much more likely to end up in the ground state than in the first excited state.

As a check on our calculations we can make sure that these values do not sum up to more than one. Upon checking we find that

$$P(|\psi_0(2\omega_0, x)\rangle) + P(|\psi_2(2\omega_0, x)\rangle) \approx 0.9952$$

which confirms that our calculations are at least close. This also implies that the particle only has a 0.48% chance of ending up in a state other than the ground state or second excited state.

Part c

Since this perturbation is applied on a much longer time scale than the natural time scale of the system, the particle will end up in the ground state of the perturbed potential. The wavefunction of this perturbed ground state is

$$|\psi_0(2\omega_0, x)\rangle = \left(\frac{2m\omega_0}{\pi\hbar} \right)^{\frac{1}{4}} \exp\left(-\frac{m\omega_0 x^2}{\hbar}\right).$$

Hence, the overlap with the perturbed ground state is 1.

$$\boxed{P(|\psi_0(2\omega_0, x)\rangle) = 1}$$

Part d

As with part c, the slow perturbation will cause the particle to end up in the new ground state with 100% probability. It will therefore have no overlap with the excited states.

$$\boxed{P(|\psi_1(2\omega_0, x)\rangle) = 0} \quad \& \quad \boxed{P(|\psi_2(2\omega_0, x)\rangle) = 0}$$